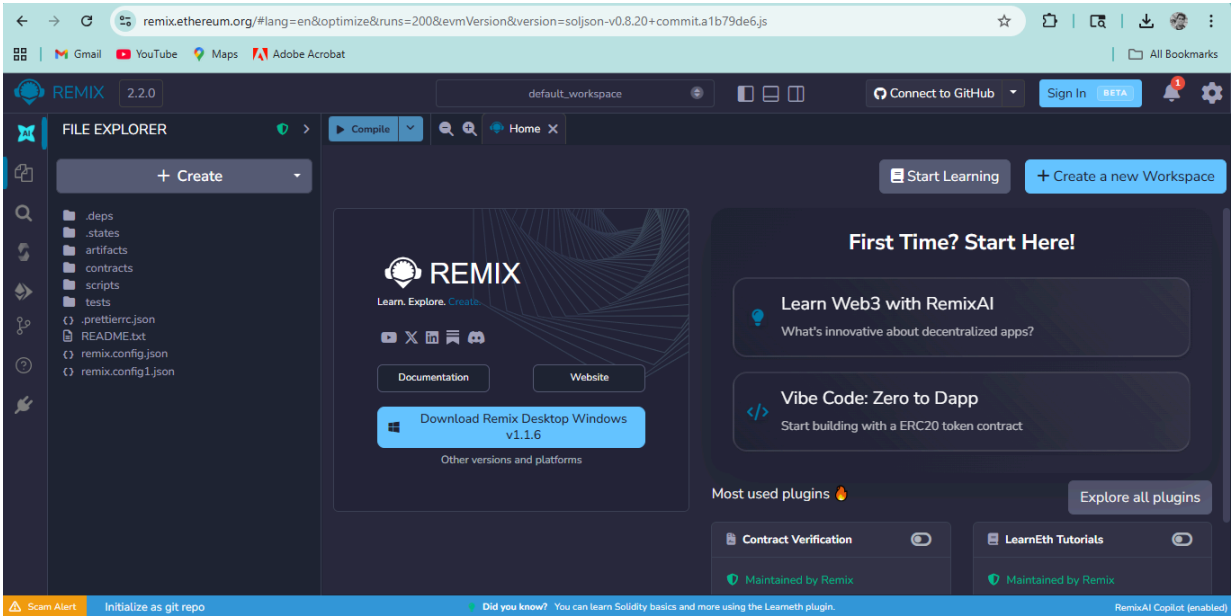


Practical 9

Step 1: Open Remix IDE

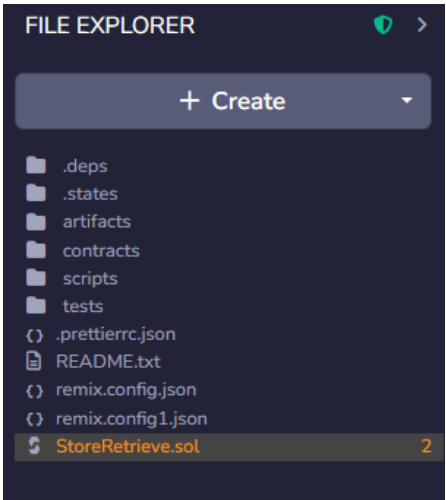
- Open browser (Chrome/Firefox)
- Go to: <https://remix.ethereum.org>
- This provides editor + compiler + blockchain simulator



Step 2: Create a New File

- Click File Explorer
- Click New File (+)

Name it: StoreRetrieve.sol



Step 3: Write Smart Contract Code

- Write a simple contract to store and retrieve a number
- Include:
 - storedNum (state variable)
 - store() function → saves value
 - retrieve() function → returns value

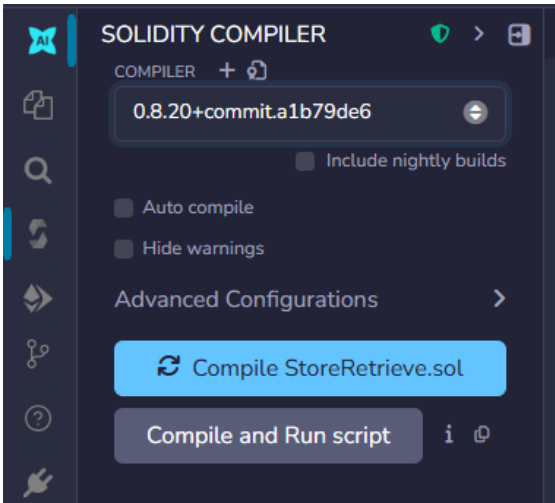
▶ Compile

StoreRetrieve.sol

```
1 //SPDX-License-Identifier: UNLICENSED
2 pragma solidity ^0.8.20;
3
4 contract StoreRetrieve {
5     uint public storedNum;
6
7     //Store a number on the blockchain
8     function store(uint number) public{ 22536 gas
9         storedNum = number;
10    }
11
12    //Read the stored number
13    function retrieve() public view returns(uint){ 2410 gas
14        return storedNum;
15    }
16 }
```

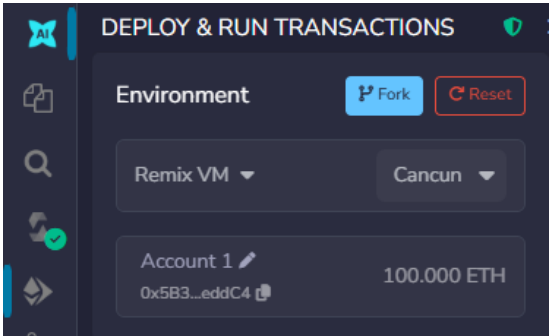
Step 4: Compile the Contract

- Go to Solidity Compiler
- Select version 0.8.20
- Click Compile StoreRetrieve.sol
- Ensure no errors (✔)



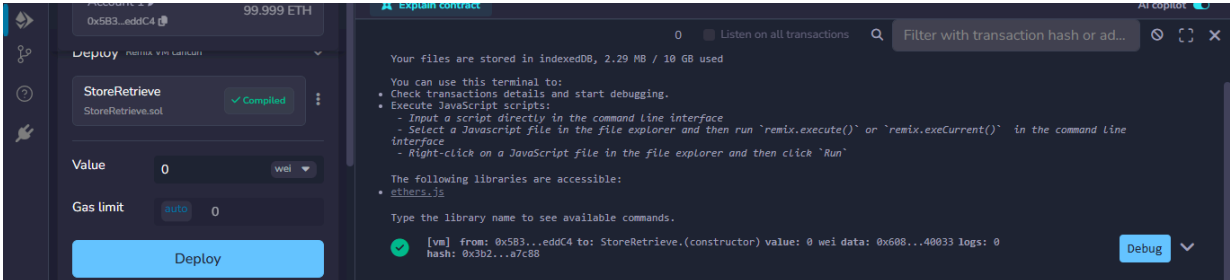
Step 5: Select Environment

- Go to Deploy & Run Transactions
- Choose Remix VM (Cancun/Shanghai)
- This creates a local blockchain



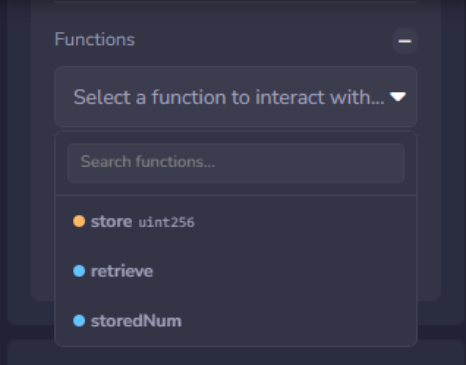
Step 6: Deploy the Contract

- Click Deploy
- Contract gets deployed with an address



Step 7: Execute Functions

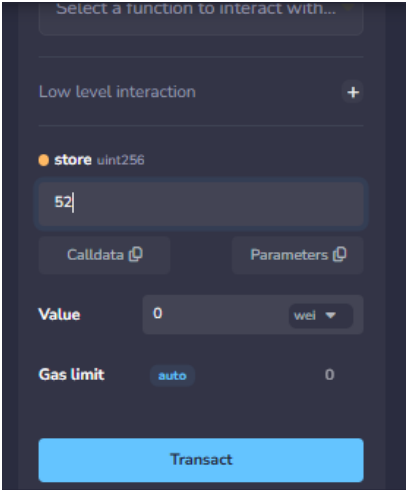
- Expand Deployed Contracts



- Use available functions:

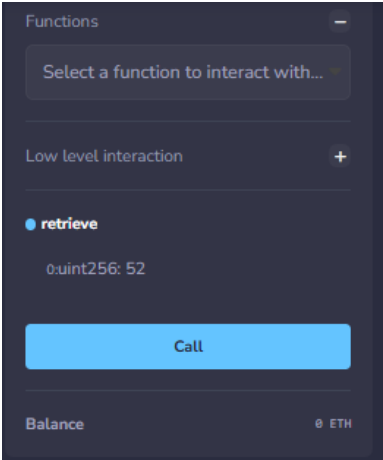
- **7.1 Store Value**

- Enter a number (e.g., 52)
- Click store → Transact
- Value is saved on blockchain



- **7.2 Retrieve Value**

- Click retrieve → Call
- Stored value is displayed



Result

- Data is successfully stored and retrieved from blockchain
- Smart contract executed successfully